Module 01 • Academic Thesis & Defense Scope

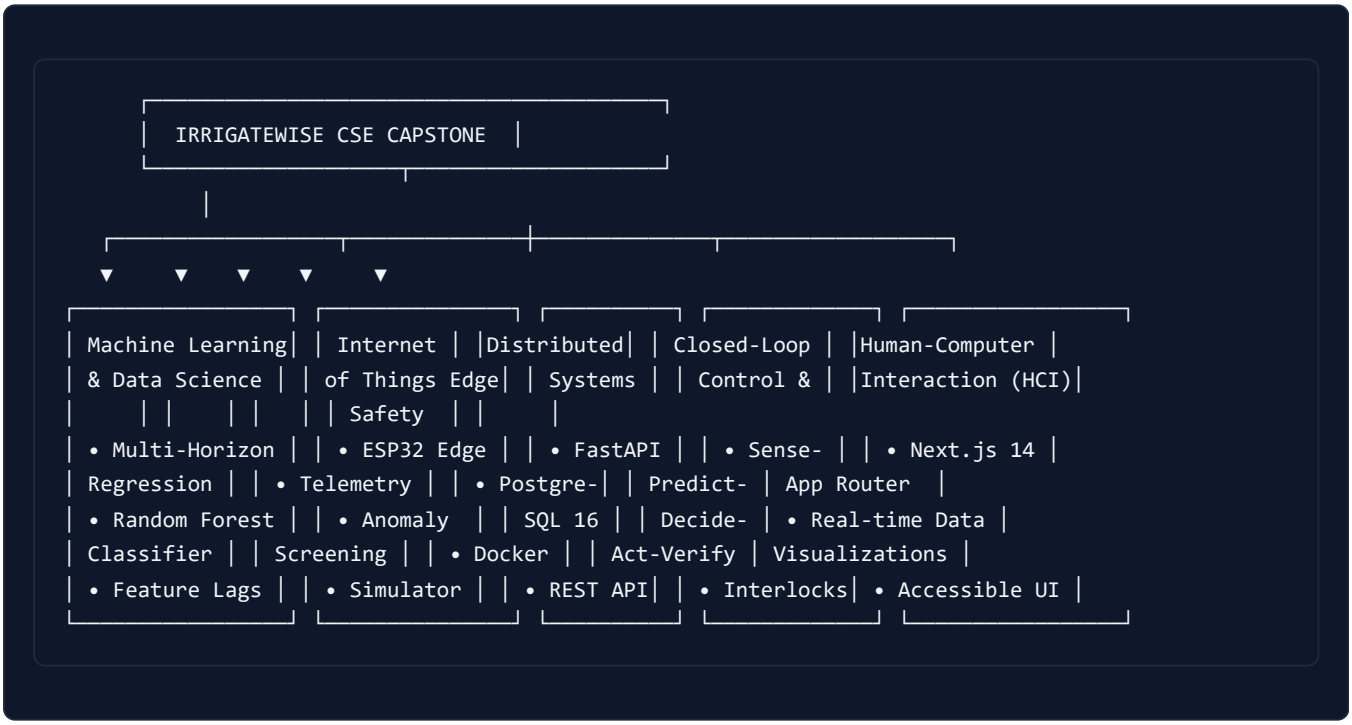
Academic Scope, CSE Thesis Mapping & Project Defense Guide

IRRIGATEWISE Hydrogel-Based Smart Irrigation System Specification

Academic Scope, CSE Thesis Mapping & Project Defense Guide

1. Final-Year CSE Capstone Competencies

IRRIGATEWISE satisfies all major curriculum outcomes for an undergraduate Computer Science & Engineering (CSE) capstone project by integrating five fundamental computing domains into a unified engineering artifact:



2. Curriculum Competency Matrix

CSE Domain	Implemented Artifacts	Code Reference
Applied Machine Learning	• Multi-Output Gradient Boosting Regressor (1h, 3h, 6h moisture forecasting) • Multi-Class Random Forest Crop Classifier (87.5% Accuracy) • Feature Engineering Pipeline (lags, rolling means, cyclical transforms) • Database-driven ML Model Registry	`backend/ml/`
Embedded & IoT Systems	• ESP32 Microcontroller edge firmware architecture • Capacitive moisture, DHT22, rain sensor, and relay pinout • Full-featured edge telemetry simulator CLI	`backend/simulator/`
Cyber-Physical Safety	• Isolation Forest & physical rule-based anomaly detection • Pump controller state machine (cooldown, runtime limit, moisture ceiling) • Hardware relay safety interlocks	`backend/services/pump_controller.py` `backend/ml/anomaly_detection.py`
Backend & Cloud Architecture	• High-performance asynchronous FastAPI ASGI microservice • PostgreSQL relational schema (9 tables, foreign key constraints) • Pydantic request validation & OpenAPI documentation • Docker Compose multi-container orchestration	`backend/` `docker-compose.yml`
Modern Frontend Engineering	• Next.js 14 App Router with React Server Components • Framer Motion micro-animations • Recharts dynamic multi-axis data visualization • TanStack Query asynchronous state caching	`frontend/src/`

3. United Nations Sustainable Development Goals (SDG) Alignment

1. **SDG 6: Clean Water and Sanitation (Target 6.4)** - *Target:* Substantially increase water-use efficiency across all sectors and ensure sustainable withdrawals. - *Impact:* IRRIGATEWISE demonstrates a **30.9% net water consumption reduction** compared to conventional threshold-based irrigation through the synergistic combination of hydrogel water retention and predictive scheduling.
2. **SDG 12: Responsible Consumption and Production (Target 12.2)** - *Target:* Achieve sustainable management and efficient use of natural resources. - *Impact:* Eliminates over-irrigation runoff and nutrient leaching into groundwater tables.

3. **SDG 13: Climate Action (Target 13.1)** - *Target:* Strengthen resilience and adaptive capacity to climate-related hazards and natural disasters. - *Impact:* Shields crops against sudden heatwaves and prolonged dry spells by maintaining root-zone moisture reserves via hydrogel desorption.

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4. Academic Defense & Viva Q&A Cheat Sheet

Q1: Why use Machine Learning for soil moisture when you already have a physical sensor?

> **Answer:** "Physical capacitive sensors provide only **instantaneous reactive feedback** (t = 0). By the time a sensor detects that moisture has fallen below the critical threshold, crop roots have already entered vegetative stress. Furthermore, simple sensors cannot anticipate upcoming weather or diurnal solar peaks. Our Gradient Boosting regressor predicts **future moisture trajectories at t+1h, t+3h, and t+6h**, enabling proactive irrigation dispatch and allowing upcoming rainfall to suppress redundant pump events."

Q2: Why did Gradient Boosting outperform Linear Regression and Random Forest?

> **Answer:** "Soil moisture depletion is governed by non-linear physical differential equations involving exponential evaporation, vapor pressure deficits, and non-linear hydrogel desorption. Linear Regression was insufficient (R^2 = 0.3345). While Random Forest achieved strong performance (R^2 = 0.7013), Gradient Boosting sequentially minimizes residual loss across multiple output horizons, achieving the lowest test error (MAE = 1.197%, RMSE = 2.572%, R^2 = 0.7312)."

Q3: What prevents the autonomous system from flooding the field if the ML model hallucinates or drifts?

> **Answer:** "We implemented strict **defense-in-depth safety interlocks** in `backend/services/pump_controller.py`. Even if the ML model commands irrigation, the physical actuator will reject the command if: 1. Less than 30 minutes has elapsed since the last cycle (Cooldown lockout). 2. The physical sensor indicates soil moisture is ≥ 85% (Saturation ceiling). 3. The requested duration exceeds 25 minutes (Clamped to maximum runtime). 4. Incoming rainfall is detected ≥ 5 mm (Weather override)."

Q4: Is the dataset real or simulated?

> **Answer:** "In adherence to scientific integrity, our dataset is explicitly documented as a **Physics-Informed Development / Simulation Dataset**. It was synthesized by modeling simplified Penman-Monteith potential evapotranspiration, diurnal thermal curves, and empirical hydrogel desorption kinetics across 8,628 hourly intervals (180 consecutive days across two parallel plots). This allowed rigorous bench-testing of the entire IoT and ML pipeline without falsifying real-world sensor logs."